

# Change Log

Measurement of  $\nu_\mu + \bar{\nu}_\mu$  Charged-Current Single Pion Production on Argon in the NuMI Beam

Corrections that changed a result

Draft 1.5, 2026-09-06

release 2026-09-06 · 51 differential extractions + 6 one-bin totals · blind

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## Abstract

This document records every correction that materially moved a number in the  $\text{CC}\pi^\pm$  NuMI analysis, with the value before and after. It exists so that a reader of the analysis note never has to work out whether a given table predates a given fix, and so that a result quoted from an older draft can be identified as superseded rather than merely different.

Most entries are normalisation errors. That concentration is itself worth stating: the extraction divides by four numbers — the data exposure, the integrated flux, the argon target count, and the per-run Monte Carlo exposure — and eight separate defects came from one of those four differing between configurations without anything detecting it. The response is `norm_manifest.py`, which reads the authoritative configuration files and asserts that they agree, so that a future drift fails a check instead of producing a wrong cross section.

Three corrections changed physics conclusions rather than only normalisation, and the conclusions they invalidated are marked as withdrawn here and in the analysis note: the non-uniform-grid regularisation defect, which had collapsed several additional-smearing matrices towards rank one; the beam-axis convention, under which reconstructed angular and transverse observables were not the same quantities as the generator predictions they were being compared against; and the double application of  $A_C$  to the model predictions, which invented a muon-angle anomaly.

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## 1 How to read this document

Each entry states what was wrong, how it was found, what changed numerically, and what — if anything — had to be withdrawn. “Before” values are quoted from the draft in which they appeared and should not be used; they are given so that an older table can be recognised.

Entries are ordered by the stage of the analysis they affect, not chronologically: exposure and flux first, then the Monte Carlo weighting, then the unfolding, then the systematic covariance.

## 2 Exposure, flux, and target normalisation

### 2.1 Obsolete integrated flux (factor $\sim 3.48$ )

An outdated flux file was used to convert the extracted event rate into a cross section. The integrated flux it supplied was larger than the correct value by a factor of approximately 3.48, so every cross section produced with it was smaller than the truth by the same factor.

**Effect.** All cross sections, all configurations. Any absolute normalisation from a draft predating this fix is wrong by  $\sim 3.48$  and must not be quoted. Shapes were unaffected, because the error was a single multiplicative constant.

### 2.2 Configuration-specific flux was not being used

The RHC and combined extractions were normalised with the FHC integrated flux. The three exposures see genuinely different spectra, and the correct integrated fluxes are

$$\Phi_{\text{FHC}} = 6.81159 \times 10^{-10}, \quad \Phi_{\text{RHC}} = 6.44646 \times 10^{-10}, \quad \Phi_{\text{comb}} = 6.60865 \times 10^{-10}$$

in  $\text{cm}^{-2} \text{POT}^{-1}$ .

**Effect.** RHC cross sections rose by 5.7% and combined by 3.1%. FHC was already correct and did not move. The flux is now set by an explicit `Flux` line in each cross-section configuration file rather than inherited, and `norm_manifest.py` checks that all configurations of a given exposure agree on it.

### 2.3 Per-mode rather than per-run exposure weighting

Data and Monte Carlo exposures were pooled per horn mode and a single scale factor applied. Because the data-to-Monte-Carlo exposure ratio differs substantially between runs, this weighted the result by how much simulation happened to exist for each run rather than by the data actually collected. Each run is now scaled by its own  $S_r = \text{POT}_r^{\text{data}} / \text{POT}_r^{\text{MC}}$ .

**Effect.** Changed the relative weight of runs within every configuration. This is also the change that makes the framework ready for real data, where the per-run exposures are not free parameters.

### 2.4 Duplicated run-total POT in split files

Two conventions coexist in the processed samples. In `processed/` a run split across several files repeats the *run-total* exposure in every file; in `processed/w/` each file carries its own *true* exposure. Dividing by the per-file value broke the first convention and summing over files broke the second: the first made the inclusive tune come out a factor of two low, the second drove an RHC proton-tagged cross section negative.

The resolution is to normalise by the sum of *distinct* per-run exposures, which is correct under both conventions.

**Effect.** At the time of the fix the RHC  $W_{\pi p}$  cross section moved from  $-0.194$  to  $+0.578$  (in units of  $10^{-38} \text{cm}^2/\text{Ar}$ ), and the inclusive RHC  $p_\mu$  result was unchanged at 0.860. Results

retracted on the strength of the negative value were reinstated. Both numbers have since moved again under the central-value weighting fix below; the current values are in the analysis note.

### 3 Monte Carlo weighting

#### 3.1 Central-value weight applied twice to the fake data

This is the largest single correction and it affected every result.

The fake data are generated externally by `throw.perrun*.C`, which folds the tuned central value, the PPFX central value and the normalisation into the event multiplicity and then writes the per-event weights back as unity — the throw is a Poisson realisation of an already-weighted expectation. The framework’s `build.universes` nevertheless substituted the *weighted* central-value histogram for the raw counts, on both the reconstructed and the truth side, so the central value was applied a second time.

**How it was found.** By asking directly whether the unfolded result was being compared against the input that generated it. Four independent checks confirmed the diagnosis: 1621.0 is a direct count of selected events across the four FHC files; 1721.9 equals both `FakeDataMC_0_reco` and `onBNB_reco – extBNB_reco`; run 1 gives 558.0 entries unweighted against 599.9 weighted for the same 558 events; and the throw’s own expectation of 1567.1 agrees with the thrown count at  $1.3\sigma$ .

**Effect.** The mean inclusive closure ratio moved from 1.106 to 1.001 (1.016 once the truth-side correction below is included, 1.018 on the re-thrown fake data), and all 51 extracted cross sections fell by between 3.8% and 12.9%. Every absolute cross section from a draft predating this fix is high by roughly a tenth.

**Withdrawn.** Two earlier explanations of the closure excess — one attributing it to the unfolding covariance and Wiener filter, one to  $A_C$  smearing — were both wrong and are retracted. They are retained in the technical supplement as recorded dead ends rather than deleted.

#### 3.2 Generator prediction histograms truncated at the overflow

Generator predictions were read without their overflow contents, so the highest-momentum region was silently missing from the comparison curves.

**Effect.** Model comparison only; the measured cross sections were unaffected. Any generator  $\chi^2$  from a draft predating the regeneration is invalid.

### 4 Unfolding

#### 4.1 Second-derivative regularisation on a non-uniform grid

The Wiener-SVD second-derivative regulariser used the uniform-grid  $(1, -2, 1)$  stencil on bins whose widths differ by factors of ten to fifteen. On such a grid that operator is not an approximation to the second derivative of anything, and its near-null direction is not the physically flat one. The Wiener filter can then retain that direction alone, which drives the additional-smearing matrix  $A_C$  towards rank one and destroys the shape information the measurement is supposed to carry.

The corrected operator differences the *density* on a non-uniform grid, reducing to the old stencil when the widths are equal, with one-sided first-derivative rows at the ends so that the operator stays non-singular without asserting that the distribution vanishes at the edges — which it does not, since  $\cos\theta_\mu$  peaks at the upper edge.

**Effect** on the conditioning measure  $s_2/s_1$  of  $A_C$ :

| Observable                | before | after |
|---------------------------|--------|-------|
| FHC $\theta_\mu$          | 0.015  | 0.51  |
| combined $\theta_\mu$     | 0.045  | 0.38  |
| proton-tagged $W_{\pi p}$ | 0.022  | 0.60  |
| proton-tagged $p_n$       | 0.21   | 0.88  |

**Withdrawn.** The earlier conclusion that  $\theta_\mu$  “repairs” what  $\cos\theta_\mu$  cannot is retracted: under the defective operator every model was being collapsed onto a common shape, so the apparent agreement was an artefact of the regularisation rather than a property of the data. After the fix  $\theta_\mu$  discriminates between models again and prefers the tune that generated the fake data.

This entry is the clearest illustration of why closure alone is not a sufficient validation. A rank-one extraction closes perfectly against its own correspondingly smeared truth while retaining almost no shape information.

## 4.2 Angular and transverse observables were defined about detector $z$

The NuMI neutrinos arrive approximately  $28.6^\circ$  away from the detector  $z$  axis, but reconstructed angles and transverse kinematic imbalance were being computed about  $z$  while the generator predictions were computed about the neutrino direction. These are not the same observables. Truth-level quantities now use the per-event neutrino direction and reconstruction uses the target-to-vertex direction; the two constructions differ only at the milliradian level.

**Effect.** 64.6% of signal events change  $\cos\theta_\mu$  bin and 44.6% change  $\cos\theta_\pi$  bin. The mean  $\delta p_T$  moves from 0.742 to 0.264 GeV/ $c$ . The muon and pion momenta, the opening angle  $\theta_{\mu\pi}$ , the hadronic mass  $W_{\pi p}$ , and the event selection are all unaffected.

**Withdrawn.** All proton-tagged transverse-kinematic model comparisons made before this correction are withdrawn, because the measured and predicted quantities were different observables.

## 4.3 Predictions were smeared by $A_C$ twice

Every model prediction plotted or compared against the data was multiplied by the additional-smearing matrix twice: once by the extractor, which applies  $A_C$  to each entry of the prediction map in place, and again on the plotting path. Confirmed exactly, as  $A_C^2 \cdot \text{raw} = \text{plotted} \times \text{width}$  to 1.0000 in every bin.

**How it was found.** By testing whether the released  $A_C$  lets an external user reproduce a published generator curve. It did not — the discrepancy was 31% on the integral and varied from 1.21 to 1.55 across bins, so it was not a normalisation — and the residual factor turned out to be  $A_C$  itself.

**Effect** (2026-08-31 pass; the current values are in the note’s  $\chi^2$  tables). Model  $\chi^2$  values only. The anomaly the note previously attributed to  $A_C$  suppression of sharply peaked generators was this bug:

| Observable           | Model | before  | after  |
|----------------------|-------|---------|--------|
| FHC $\theta_\mu$     | NEUT  | 40.60/5 | 8.96/5 |
| FHC $\theta_\mu$     | NuWro | 38.62/5 | 9.43/5 |
| FHC $\cos\theta_\mu$ | NEUT  | 29.78/5 | 4.90/5 |
| FHC $\cos\theta_\mu$ | NuWro | 29.10/5 | 4.93/5 |
| FHC $p_\mu$          | GENIE | 5.52/7  | 0.57/7 |

**Not affected.** The unfolded cross sections (FHC  $p_\mu$  integrates to 0.8275 before and after) and the closure test, because the fake-data truth was the one curve not passed through the

second multiplication. The previous table also predated the central-value weighting fix, so part of the shift in its “truth” column is that correction rather than this one.

**Withdrawn.** The argument for preferring  $\theta$  over  $\cos\theta$  on the grounds that the  $\cos\theta_\mu \chi^2$  was an artefact. The effect that motivated it does not exist.

## 5 Systematic covariance

### 5.1 Combined detector-variation exposure double-counted

The combined configuration pooled detector-variation samples per horn mode, and the denominator used a single file’s exposure rather than the summed Monte Carlo exposure over all files of that variation type, so the combined detector systematic was inflated.

**Effect.** The total detector systematic fell from 42.3% to 24.6%, and the combined result moved from 0.910 to 1.027 times its expected value. FHC and RHC were bit-identical before and after, which is the check that isolates the defect to the combination step.

### 5.2 Proton-tagged detector systematics were absent entirely

Of the fifty-one extractions, the thirty-three proton-tagged ones carried *no* detector systematic at all: the `detVar` column was `n/a` for every proton-tagged observable in every configuration, so their quoted uncertainties omitted a term worth 17–43% in the corresponding inclusive results.

The cause was a configuration gap rather than missing data. The proton-tagged detector variation samples had been processed on 2026-08-20/21 and carry the required branches, but nothing referenced them: the `_w` file-properties tables contained no `detVar` lines, and `ccpi1p_syscalc_numi.conf` had no DV block at all. Both were needed; adding either alone does nothing.

**Effect** (values of the time). The thirty-three proton-tagged universes were rebuilt with the nine detector variation samples included and re-unfolded from them. Every extraction in the analysis now carries a detector term. The proton-tagged terms ran 10.5%–64.4% with a median of 23.9%, against 16.9%–42.8% for the inclusive results (current release: 10.5–108.5%, median 33.0%, and 15.6–35.0%).

**Check** (of the time). Per-mode detector-variation double counting has occurred in this analysis before, and the combined configuration — which carries two per-mode sets where the single-mode configurations carry one — is the one exposed to it. Double counting inflates the combined result. On that pass, of the eight observables (of eleven) whose combined value fell outside its own FHC–RHC range, six fell *below* both and only two above, the finite-sample averaging-down expected of statistically limited detector-variation samples rather than a recurrence; on the current release only two of eleven fall outside the range, both above (analysis note).

### 5.3 Cut-and-count uncertainty quoted from a pre-rebuild breakdown

The total flux-averaged cross sections of the cut-and-count path took their systematic fraction from the  $p_\mu$  systematic-breakdown tables, which had not been regenerated after the detector-variation rebuild of 2026-08-31 (their detector terms were  $\sim 10\%$  against 17–43% in the release). The breakdown tables in the note and the supplement were regenerated from the release systematics dumps on 2026-09-02 and the cut-and-count uncertainties recomputed from them; the macro also now applies the central-value MC weight, which moved the central values by 0.3–0.5%.

**Effect** (values of the time; the current totals are in the one-bin entry below). Inclusive totals ( $10^{-38} \text{ cm}^2/\text{Ar}$ ): FHC  $1.06 \pm 0.27 \rightarrow 1.03 \pm 0.43$ , RHC  $0.98 \pm 0.35 \rightarrow 0.93 \pm 0.42$ , combined  $1.02 \pm 0.30 \rightarrow 0.98 \pm 0.40$ . The generator comparison changes from “GENIE  $\sim 1\sigma$  low” to

“GENIE 0.5–0.6 $\sigma$  low”. A proton-tagged cut-and-count total is quoted for the first time (FHC  $0.63 \pm 0.31$ , RHC  $0.55 \pm 0.21$ , combined  $0.59 \pm 0.25$ ).

#### 5.4 Fake-data truth still carried the central-value weight

The 2026-08-30 correction (above) stopped the tune weight being applied a second time to the externally thrown fake data on the reconstructed side. The truth side was missed: the throw macro resets only the scalar weight branches, the `weight_TunedCentralValue_UBGenie` vector survives, so the universe maker still writes a tune-weighted “CV” universe for each fake-data file, and the systematics calculator took that weighted histogram as the fake-data *truth* whenever it was present. Measured on the FHC  $p_\mu$  universe file: weighted/unweighted true-signal integrals 1.023, 1.008, 1.008, 1.018 for Runs 1, 2, 4, 5, about 1.6% overall. Every closure ratio was therefore low by that amount and every closure  $\chi^2$  slightly too small. Found in the 2026-09-02 code trace; the truth path now uses the same external-throw flag as the reco path, and the universe cache key was bumped so the stale sums cannot be reused.

**Effect.** Unfolded cross sections and covariances are unchanged (FHC  $p_\mu$  integrates to 0.8275 before and after). Closure: FHC  $p_\mu$  unfolded/truth  $1.024 \rightarrow 1.041$ ,  $\chi^2$   $0.10/7 \rightarrow 0.13/7$ ; the mean inclusive closure ratio  $1.001 \rightarrow 1.016$ . All fifty-one sidecars, the released curves (`truth_smeared` column), the closure tables and the closure figures were regenerated; the ensemble bias is unaffected because its reference is the fixed central-value truth, not the thrown one.

#### 5.5 Beam-off gate count was an event count

Every live file-properties configuration scaled the pooled Run-1+Run-3b beam-off sample by `bnb_trigs[run]/3821593`. The denominator was inherited from the earlier custom selection, where it was the sum  $610\,496 + 3\,211\,097$  of the two samples’ *event* counts; their gate counts, from the analyser’s trigger table supplied on 2026-09-02, are  $4\,582\,248.27 + 32\,649\,128.65 = 37\,231\,376.92$ , a factor 9.74 larger. The selected cosmic rate per gate exposed it: 5.8 per  $10^6$  gates for the pooled sample against 0.37–1.3 for the dedicated per-period files, versus 0.59 with the gate count. Corrected in all six live configurations and in the counting and cut-flow macros; the universes were re-summed (the cache key tracks the file-properties contents) and all fifty-one extractions, covariances and the release were regenerated.

**Effect.** EXT background at the FHC final cut  $128 \rightarrow 13$  events (RHC  $133 \rightarrow 14$ , combined  $261 \rightarrow 27$ ); purity  $55.8 \rightarrow 59.7\%$  (FHC). Fake-data central values are unchanged by construction (the same EXT is added and subtracted); the EXT-statistics and data-statistics covariance terms shrink, which moves the Wiener-SVD results only through the regularisation (see the regenerated tables). A real-data extraction with the old value would have over-subtracted  $\approx 115$  events,  $\approx 12\%$  of the FHC signal. **Withdrawn:** every beam-off yield, purity and EXT-statistics term quoted in drafts before 1.2. The sideband tables were regenerated with the per-run treatment (`regen_sidebands.sh`); the cosmic control region, formerly EXT-dominated by  $4.5 : 1$ , now has 0.7 EXT events per neutrino-MC event.

#### 5.6 Beam-off matched by run period instead of one pooled sample

Beam-off data are cosmic-only, so a sample is matched to a beam-on run by its run period, irrespective of the horn polarity it was recorded under (analyser, 2026-09-02). The pooled Run-1+Run-3b sample was replaced by the dedicated per-period samples (Run 1; Run 3b; Run 4a, 4b, 4c, 4d pooled; Run 5; the Run-1 sample standing in for Run 2), each scaled by its own gate count from the analyser’s table, and processed with the `swtrig` fail-open build so that the CRT-era samples select cosmuics at all. The selected rate per  $10^6$  gates is 1.75 (Run 1), 0.43 (Run 3b), 1.15 (Run 4), 0.52 (Run 5), so period matching moves the prediction by a factor of about two relative to the pooled sample. All fifty-one universe files were rebuilt (SLURM array 3205007, 51/51), the extractions re-unfolded, covariances re-harvested and the release regenerated.

**Effect.** EXT background at the final cut (FHC / RHC / combined): pooled 13/14/27  $\rightarrow$  per-run 30/23/53 events, i.e. 1.9% of the selected sample; purity 59.1% (FHC). EXT-statistics term on FHC  $p_\mu$  0.4  $\rightarrow$  0.9%. Unfolded central values move by  $\lesssim 1\%$  through the regularisation.

## 5.7 PPFX central-value weight absent from the cut-flow and cut-and-count counts

The framework’s central-value event weight for NuMI is  $\text{tune} \times \text{PPFX CV} \times \text{normalisation}$  (`UniverseMaker.cxx`), and the per-run fake data are thrown with that product. The selection’s cut-flow histograms and the cut-and-count macro carried the tune only. The PPFX central-value weight averages 0.944 on selected signal, so every absolute yield in the cut-flow table and every cut-and-count total sat 3–6% above the prediction the unfolding uses; efficiencies and purities, being ratios, were unaffected at the 0.5% level. Found in the 2026-09-02 code trace.

**Effect.** Cut-and-count totals: FHC 1.06  $\rightarrow$  1.03, RHC 0.98  $\rightarrow$  0.93, combined 1.02  $\rightarrow$  0.98 ( $10^{-38}$  cm<sup>2</sup>/Ar), and the RHC/FHC ratio 0.928  $\rightarrow$  0.901. The selection now fills the cut-flow histograms with the full weight and the fourteen overlay files and the dirt overlay were reprocessed on 2026-09-03 (`reprocess_cutflow_cf.sh`, into a separate directory so the universes’ POT convention is untouched); the cut-flow table is regenerated from them. No unfolded result is affected.

## 5.8 Proton-tagged TKI figure drawn from the withdrawn fine-binned sidecars

The proton-tagged W/TKI montage macros read closure sidecars keyed `dpt`, `dalphot`, `dphit`, `pn`: the withdrawn fine-binned extraction of 2026-08-18, which predates the beam-axis correction, rather than the released two-bin `*2bin` files. The figure therefore compared transverse quantities about detector  $z$  (its “data” and tune) with beam-frame generator curves, and showed shape disagreements of a factor of several that do not exist in the release. The staleness check maps figures by configuration only and could not catch it. Macros repointed, the stale sidecars quarantined, the figure regenerated (2026-09-03).

**Effect.** Figure only; no released number. In the two-bin scheme the generators differ by normalisation ( $\times 1.6$  GENIE to NEUT) with shapes consistent with the data.

## 5.9 Result tables rewritten into the wrong place by the table patcher

The script that repatched the result tables after each regeneration located a table by the `\begin{tabular}` nearest its label, falling back to the *preceding* tabular whenever the caption between label and tabular exceeded 400 characters. Four tables were affected in the committed Draft 1.2: the FHC results table received the RHC rows, the RHC table the combined rows, the combined table the rows of the integrated-cross-section summary, and in the supplement the cut-and-count comparison was overwritten by an FHC systematics breakdown. Caught by the second external review (2026-09-03). All release-derived tables are now regenerated as whole tabulars by `rebuild_tables.py` with a locator that requires the tabular to follow the label inside the same environment, and `report/check_tables.py` asserts, on every build, that each table sits under its own label, has the column count its header declares, and agrees with `current_results.tsv`.

**Effect.** Presentation only; the release files were correct throughout. The tables of the 2026-09-03 build are the first to pass the automated check.

## 5.10 Two central-value weight rules were in use; the fake data skipped events the prediction counted

The framework’s `safe_weight()` sets a non-finite, negative or  $> 30$  central-value weight to unity. The count-level macros (cut-and-count, cut-flow, sidebands, transfer factors) instead zeroed

invalid weights and did not clamp, and the per-run fake-data throws *skipped* events with a non-finite weight. About 1.15% of overlay events carry a non-finite tune weight, concentrated in the non-signal  $\nu_\mu$ CC classes, so the pseudo-data sat below the framework’s own prediction by 0.9% in the signal region (13–14 events) and by 2.8–3.4% in the CC0 $\pi$  control region (341 FHC, 422 RHC POT-scaled events); the sideband yields of the note differed from the systematics tables by the same amounts. Found through the third external review (2026-09-03).

**Correction.** One rule, the safe-weight definition in the Monte Carlo section of the note, is now applied everywhere: the selection’s cut-flow histograms (reprocessed), every count-level macro, and the fake-data throws, which were re-thrown with the same seeds and the events kept at unit weight; all fifty-one universe files were rebuilt from the re-thrown fake data. **Effect.** Pseudo-data yields rise by 0.9% (signal region) to 3.4% (CC0 $\pi$ ). Because the re-throw is a new Poisson realisation, the individual Wiener-SVD integrals moved by up to 20% (RHC  $p_\mu$  0.573  $\rightarrow$  0.690) within their statistical uncertainties, the inclusive closure mean moved from 1.016 to 1.018 (all  $p > 0.9$ ), and the prediction-total uncertainty that the cut-and-count totals inherit from the  $p_\mu$  extraction moved from 44.4% to 37.7% (RHC) and from 37.0% to 53.5% (proton-tagged RHC); the cut-and-count central values, which use the central-value prediction rather than the throw, are unchanged. The D’Agostini cross-check was re-run on the new release: its integral is 20–29% above Wiener-SVD and flat across observables to  $\leq 1.0\%$  (previously 27–33% and  $\leq 2.1\%$ ). The real-data path was never affected.

### 5.11 The total cross section borrowed its systematic from the $p_\mu$ extraction

The headline total was a cut-and-count evaluation whose systematic fraction was taken from the prediction total of the released  $p_\mu$  extraction of the same configuration. That proxy carried the regularisation amplification of the flux and detector terms, and it depended on the fake-data throw through the unfolding operator: on the re-throw of 2026-09-03 alone it moved from 44.4% to 37.7% (RHC) and from 37.0% to 53.5% (proton-tagged RHC) with no change to the central value.

**Correction.** The total is now extracted with the framework as a one-bin distribution: one true bin containing every signal event, defined on  $\cos\theta_\mu$  over its full range so that no overflow exists, one reconstructed bin, Wiener filter off. The response is the efficiency alone, nothing is regularised,  $A_C \equiv 1$  exactly, and the covariance is propagated universe by universe like every other extraction. Six new universe files (inclusive and proton-tagged  $\times$  FHC/RHC/combined; SLURM 3228639) and `data_release/total_xsec.tsv`. The cut-and-count macro is kept as the closed-form check and reproduces the one-bin extraction of the central-value prediction to four digits. **Effect.** Inclusive totals  $1.03 \pm 0.40$ ,  $0.93 \pm 0.35$ ,  $0.97 \pm 0.37$  (previously  $\pm 0.42$ ,  $\pm 0.35$ ,  $\pm 0.34$  with the borrowed systematic); proton-tagged  $0.62 \pm 0.26$ ,  $0.53 \pm 0.25$ ,  $0.57 \pm 0.24$  (previously  $0.63 \pm 0.32$ ,  $0.55 \pm 0.30$ ,  $0.59 \pm 0.28$ ). The central values now fluctuate with the throw (closure 0.98–1.00 against the realised truth) instead of being the injected tune by construction. The generator comparison moves from “0.5–0.6 $\sigma$  above GENIE, within 0.7 $\sigma$  of the rest” to “0.5–0.8 $\sigma$  above GENIE, within 0.8 $\sigma$  of the rest”.

### 5.12 Corrections from the independent review of 2026-09-04

Two reviewers with no prior contact with the analysis read Draft 1.4 in full, one checking every number against the release files and one as a referee. What they found, and what changed:

**Release index.** `data_release/index_extractions.tsv` and `README.md` were the 2026-08-31 files and disagreed with every other release artefact; regenerated from the current results (`report/tools/index_extractions.py`), with the differential  $W_{\pi p}$  shape flagged withdrawn.

**Proton-tagged cut-flow.** The supplement’s proton-tagged cut-flow table predated the beam-off gate and PPFX-weight corrections, and the note’s proton-tagged prose quoted it (47–48% purity, 33  $\rightarrow$  48%). The fourteen overlays and the dirt overlay were reprocessed with the

proton-tagged selection under the unified weight rule (`processed/cf_1p`); the macro’s final stage, which had counted events unweighted (2.6% high), now carries the same weight as the other stages, and its final row reproduces the cut-and-count check exactly (376.4 signal, 723.0 predicted, FHC). Purity is 52% (FHC), 50% (RHC), 51% (combined); the tag rejects 67% of the neutrino background for a 32% signal loss and reduces the beam-off by 3.3–3.5 at the final cut, not six.

**Flux term.** The note attributed the 24–33% unfolded flux term to amplification by the unfolding operator. The one-bin total, which has no regularisation, carries 28–30%: a flux universe scales signal and background together, so the term is  $\delta\Phi/\Phi \times (1 + B/S)$ ,  $17.7\% \times 1.71 = 30.2\%$  against the one-bin 30.25% (FHC). The section is rewritten; the reconstructed-level table of the earlier pass is dropped.

**Control-region procedure.** The target class of the  $\pi^0$  region as tabulated ( $\nu_\mu\text{CC}$  with a  $\pi^0$ ) is 46% of the region, below the 50% purity the procedure requires for a fitted correction; stated, with the CC+NC  $\pi^0$  definition to be ratified. The pass/fail criterion is now a global normalisation  $\chi^2$  over the eight (region, mode) totals; the sixteen independent  $2\sigma$  and  $p > 0.05$  tests of the earlier draft (which would have flagged at least one region by chance more often than not) are diagnostics, reported with pulls against the statistical+detector covariance and a normalisation-marginalised shape test. The response to a localised reconstruction failure, the joint FHC/RHC treatment and the leverage of the held-fixed signal contamination are specified. A control-region skim that never writes signal-region events is added to the prerequisites; it does not yet exist.

**Ratified 2026-09-06.** The analysers ratified the  $\pi^0$  region’s target class as CC+NC  $\pi^0$  (purity 50.1% FHC, 51.0% RHC of the neutrino MC;  $T = 0.019\text{--}0.020$ ; transfer uncertainty 17–19%, floor-limited); the class was added to the transfer, transfer-uncertainty and detector-variation macros and to the note’s tables, and the purity denominator of the procedure is now defined as the neutrino-MC prediction.

**Protocol statistics (2026-09-06).** The three statistics of the frozen procedure (global  $8 \times 8$  normalisation  $\chi^2$ , per-region pulls against the full and the statistical+detector covariance, normalisation-marginalised shape tests) were implemented (`macros/sb_protocol.C`) and exercised on the full-stack pseudo-data:  $\chi^2$  0.66/8, pulls within  $\pm 0.5$ , shape  $p > 0.47$ ; power 12% ( $\text{CC}0\pi$ ,  $\pi^0$ ) and 50–62% (multi- $\pi$ , cosmic) for a single-region shift, 58% for a common shift. The response tree was rewritten as an explicit flow (trigger, order of regions, numeric criterion for a localised reconstruction failure, size of the added uncertainty, leverage of the held-fixed components in a fitted correction, post-treatment criterion).

**Detector term statistics.** The Monte-Carlo-statistical floor of the detector-variation term was measured on split samples (`macros/sb_detvar_split.C`): the  $\text{CC}0\pi$  transfer term (27% FHC, 31% RHC) is genuine (floor 9%, 6%); the  $\pi^0$  transfer term (15%, 11%) is at or below its floor (13%, 17%) and is an upper limit; the region-yield terms have floors below 2%. The procedure now uses the larger of the term and its floor.

**Ensemble.** The 32-throw ensemble was thrown by a macro that still skipped non-finite-weight events, which accounts for  $\approx 1.4\%$  of its 2.5% negative offset. The macro now applies the unified rule and the ensemble was re-run for  $p_\mu$ ,  $\cos\theta_\mu$  and the one-bin total (SLURM 3277674, 3277691, 3277692; statistical covariances re-harvested with `slurm/ens_statcov.sh`, pulls with `report/tools/ensemble_stat_pulls.py`). Statistics-only pull widths 1.08, 0.97, 1.02 (previously 1.04, 0.96); ensemble means below the reference by 0.5%, 2.7% and 1.0% at 0.4, 1.9 and  $1.3\sigma$  (previously 2.5% at  $2.1\sigma$  for  $p_\mu$ ). The bias statement in the note is reworded accordingly: a small common negative offset, not significant, largest in the most regularised observable, and present at the 1% level in the unregularised total.

**Also.** The  $A_C$  claims made about proton-tagged closure ratios (which are ratios to the  $A_C$ -smeared truth) are removed; the RHC-below-FHC comparison of  $A_C$ -suppressed integrals is replaced by the one-bin ratio; the reproducibility claim is scoped to the 2026-09-02 pass; the D’Agostini cross-check is stated to be prior-dominated on central-value input; the absence of

an alternative-generator fake-data test on the released chain, of an MCS momentum systematic and of gate-ratio and Run-2 stand-in beam-off terms is stated in the Limitations; the signal definition (one charged pion of any momentum, above threshold), the unfold provenance, the regularisation matrix and the proton-tag values ( $\text{LLR} < 0.05$ ,  $p_p > 0.3 \text{ GeV}/c$ ) are stated precisely; the flux mean energies (0.48 GeV) and the fraction above 0.3 GeV (40%) are given, which is the origin of the factor  $\sim 3$  against the BNB total. Some twenty further inline numbers that had survived from older passes were corrected.

### 5.13 Inclusive $\cos \theta_\pi$ re-binned to five forward-split bins

The binning re-check of 2026-09-05 (technical supplement) found that every released binning except  $\cos \theta_\pi$  is at its resolution limit, and that  $\cos \theta_\pi$  admits a fifth bin in the forward hemisphere. Adopted on 2026-09-06: edges  $-1, -0.1, 0.35, 0.6, 0.8, 1$  replace  $-1, -0.1, 0.35, 0.75, 1$  for the three inclusive extractions (the proton-tagged  $\cos \theta_\pi$  keeps four bins). The four generator predictions were regenerated at the new edges from the event-level files (`generator_predictions/costhpi5`); the unchanged observables reproduce the previous prediction files to machine precision. The three universe files were rebuilt (SLURM 3277974), all extractions re-unfolded and the release regenerated; the D’Agostini cross-check was re-run.

**Effect.** FHC  $\cos \theta_\pi$ :  $\sigma_{\text{int}} 0.726 \rightarrow 0.674$ , closure  $\chi^2 0.80/4 \rightarrow 0.68/5$ ,  $A_C$  row sums  $0.71\text{--}0.82 \rightarrow 0.68\text{--}0.77$ ,  $s_{\text{min}}/s_1 0.27 \rightarrow 0.46$ ; RHC  $0.663 \rightarrow 0.615$ ,  $0.43/4 \rightarrow 0.49/5$ ; combined  $0.753 \rightarrow 0.658$ ,  $1.05/4 \rightarrow 0.74/5$ . The inclusive closure mean moves from 1.018 to 1.010, the range of inclusive integrals to 0.62–0.96, and the D’Agostini-to-Wiener-SVD ratio to 1.23–1.31 (flat to  $\leq 1.2\%$ ). No other extraction changes.

### 5.14 Documentation-only corrections from the 2026-09-02 review

None of the following moved a released number; they are recorded because a reader of Draft 1.1 would otherwise take the printed formulas as the implementation. (i) The differential cross-section equation divided by the efficiency in addition to unfolding with the smearceptance matrix  $S = \varepsilon M$ ; the implementation applies the correction once, through  $S$ , and the equation now says so. (ii) The response-matrix figure was described as the column-normalised migration matrix; it is  $S$ , with columns summing to the efficiency. (iii) The muon-candidate cut was described as “longest qualifying track”; the code orders the qualifying set by the muon-BDT score. (iv) Coherent pion production was described as a background; the signal definition is topological and includes it. (v) The single-exposure normalisation example ( $7.63 \times 10^{20}$  POT) and a pre-rebuild systematic-breakdown table were replaced by the final three-configuration values. (vi) Flux composition was labelled as integrated above 0.1 GeV; the values are for 60 MeV. (vii) TKI angle units in the binning appendix read rad; they are degrees. (viii) A  $\theta_\pi$  result was listed as final; none is extracted. (ix) The beamline-geometry flux term, quoted at 1.6–3.3%, is not in the released systematics configuration and is now stated as absent.

## 6 Development-era corrections

These predate the framework-based analysis and are recorded for completeness. They were previously listed in a “Known Issues” table in the analysis note; that table had become thirteen resolved rows against one open one, so it was retired and its contents moved here, where a resolved correction belongs.

| Item                                                                           | Resolution                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unfolding covariance was statistics-only<br>$\theta_{\mu\pi}$ binning mismatch | Replaced by the framework <code>PredTotal</code> plus the data Poisson term, removing a $\sim 26\%$ bias in the quoted uncertainty. A custom 10-bin scheme was aligned to the framework's 9-bin one; the two pipelines had been binning the same observable differently.                                                                                                                 |
| Negative tail bins                                                             | An artefact of the EXT treatment; all bins positive once the subtraction was corrected.                                                                                                                                                                                                                                                                                                  |
| EXT subtraction on fake data                                                   | The custom pipeline cancelled it via <code>fake_data_ext_cancel</code> , a flag that had to be remembered at unblinding. Superseded: the framework selects the beam-off treatment from the presence of Monte Carlo truth in the beam-on file (added to pseudo-data, subtracted from real data) and refuses a real file unless <code>XSEC_UNBLIND=1</code> , so no manual switch remains. |
| Multisim propagation in the custom pipeline                                    | Never completed, and now moot: the released results come from the framework, which propagates the multisims ( <code>cov_xsec_multi</code> is in the release). The row survived in the note as the single “open” item long after it had ceased to describe anything.                                                                                                                      |
| CRT veto branches                                                              | <code>crthitpe</code> and <code>crtveto</code> are all-zero in the NuMI ntuples; replaced by <code>bdt_cosmic</code> , disabled at its default.                                                                                                                                                                                                                                          |

## 7 Open items

The following are outstanding as of 2026-09-06. None is a correction awaiting application; they are prerequisites, limits of the exposure, or tests not yet made. The analysis note's request and Limitations sections carry the same list.

- **Before the control regions are opened:** ratification of the frozen procedure by the conveners; the control-region skim of the beam-on files (signal-region events never written), which does not yet exist; the  $\pi^0$  definition without the final multiplicity cut, whose effect can be quantified with a reprocess of the Monte Carlo only.
- **Before the signal region is opened:** confirmation that the flux constants are the PPFX-corrected central values; a NuWro or GiBUU fake-data pass through the released configurations (no alternative-generator test exists on the released chain); the clean-cache bit-identical reproduction repeated on the 2026-09-06 release; the ensemble extended to RHC and to the proton-tagged family; the inverted-proton-PID control region for any proton-tagged result.
- **Systematic terms not in the released covariance:** a dedicated MCS muon momentum term; the beam-off gate-ratio and Run-2 stand-in terms ( $\lesssim 0.4\%$  of the total); the beamline-geometry flux term ( $\lesssim 0.2\%$ ).
- **Limits of the exposure:** the differential  $W_{\pi p}$  shape is withdrawn (no binning satisfies both the migration and the statistics requirement); the proton-tagged sample supports an integrated cross section.
- **Ensemble offset:** the ensemble means sit 0.5–2.7% below the reference at up to  $1.9\sigma$  from the same throws; the throw itself is unbiased (mean thrown selected count 0.999 of the expectation), so the offset, if real, is in the extraction; a larger ensemble would settle it.

**External reproduction of the  $A_C$ -smeared comparison — resolved.** The transformation

$$(d\sigma/dx)_i^{\text{smeared}} = \frac{1}{w_i} \sum_j (A_C)_{ij} p_j,$$

applied to a prediction supplied as a per-bin cross section, reproduces every released generator curve to a worst per-bin deviation of  $1.2 \times 10^{-10}$ ; it was the failure of this test that exposed the double- $A_C$  bug, and its success is now the regression check. No packaged reference implementation ships with the release; a user can verify their implementation against any model column of `curves_*.tsv`.

**A caveat that is not an open item but reads like one.** The closure  $\chi^2$  values have  $p$ -values clustered near unity (median 0.98 over the fifty-one extractions) because the measurement and the truth it is compared against share their systematics, so those terms cancel in the difference while still inflating the covariance used to judge it. The ensemble of throws sizes the statistical part of the band correctly; the systematic part remains untested by construction.

## 8 Superseded values that may still be encountered

| Quantity                            | superseded value           | status                                                     |
|-------------------------------------|----------------------------|------------------------------------------------------------|
| Mean inclusive closure ratio        | 1.106, 1.001, 1.016, 1.018 | now 1.010 (five-bin $\cos\theta_\pi$ release)              |
| RHC $W_{\pi p}$ cross section       | $-0.194$                   | sign was a POT-convention artefact                         |
| Combined detector systematic        | 42.3%                      | 24.6% at the fix; 21.4% ( $p_\mu$ ) in the current release |
| Combined result / expected          | 0.910                      | 1.027 at the fix (2026-08-28 pass)                         |
| FHC $\theta_\mu$ $A_C$ conditioning | 0.015                      | now 0.51                                                   |
| Mean $\delta p_T$                   | 0.742 GeV/ $c$             | now 0.264 GeV/ $c$ (beam frame)                            |
| Proton-tagged <code>detVar</code>   | n/a, then 10.5–64.4%       | now 10.5–108.5% (median 33.0%, 2026-09-04 release)         |
| Proton-tagged $\sigma$ , all obs.   | 2026-08-18 values          | superseded 2026-08-31                                      |